Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

Arthur Shune*
SkyMesa Systems Inc.

Abstract

Functional ultrasound is increasingly used in intraoperative, neonatal, and mobile brain imaging, but residual tissue motion, phase instability, and aliased flow can still generate artifactual high scores after conventional clutter suppression. We study localized matched-subspace detectors applied downstream of clutter filtering in beamformed slow-time fUS data, from a fixed band-limited statistic to locally whitened variants. On a held-out SIMUS structural benchmark, changing only the final detector on the same clutter-filtered residual sharply reduces nuisance leakage: in the intra-operative setting, nuisance false-positive rate falls from 0.998 for Kasai to 0.004 for the fixed matched-subspace statistic at matched 50% target-flow recall. In a separate held-out end-to-end benchmark, the selected fixed chains yield zero observed nuisance detections on all four held-out seed-setting pairs. On an open real-IQ rat-brain dataset, the fully whitened variant improves a conservative localization-derived vessel-core versus shell endpoint in all 10 audited blocks (AUC 0.530 versus 0.504 for power Doppler; exact paired sign-flip $p = 0.002$). These results show that the downstream detector is a consequential design choice in post-beamformed fUS, and that localized matched-subspace scoring can reduce nuisance leakage on unchanged clutter-filtered residuals.

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone, but that even a few artifactual false detections can persist in the Doppler map and complicate interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

In intra-operative settings, a false-positive activation region can mislead surgical planning. In resting-state connectivity studies, a single artifactual hot spot that survives across blocks can create a spurious seed correlation. In neonatal bedside monitoring, where acquisition windows are short and motion is unavoidable, artifact leakage in the Doppler map can be hard to distinguish from genuine hemodynamic variation without independent confirmation that is rarely available. These are not average-sensitivity problems; they are rare-event problems driven by the right tail of the score distribution.

*Correspondence: `arthur@skymesasystems.com`. ORCID: `0009-0004-6433-3856`.

This problem matters now because fUS is no longer only a preclinical proof-of-concept modality. It is already used in intra-operative, neonatal, cranial-window, and emerging mobile settings [1, 8, 9, 21–23], where rare nuisance-driven false positives can matter more than average image quality.

On the held-out SIMUS-Struct benchmark, changing only the downstream detection statistic sharply reduces nuisance leakage on the same clutter-filtered residual. In the intra-operative setting, nuisance FPR falls from 0.998 for Kasai to 0.004 for the fixed matched-subspace statistic at matched 50% target-flow recall. In the separate best-chain benchmark, the selected fixed detector chains yield zero observed nuisance detections on all four held-out seed-setting pairs.

After clutter filtering, we replace the final PD/Kasai-style readout with a local matched-subspace score that asks whether residual slow-time energy is concentrated in the expected flow band. We study three variants: a fixed statistic, an adaptive version that invokes whitening in clutter-dominant neighborhoods, and a fully whitened version. Across the settings studied here, the fixed statistic is the default structural choice, whereas whitening helps in more clutter-dominant settings.

Figure 1 gives a qualitative real-IQ rat-brain preview using the fully whitened variant, which is strongest on this ULM structural endpoint. The default detector choice for general structural use is determined separately by the held-out SIMUS benchmark below.

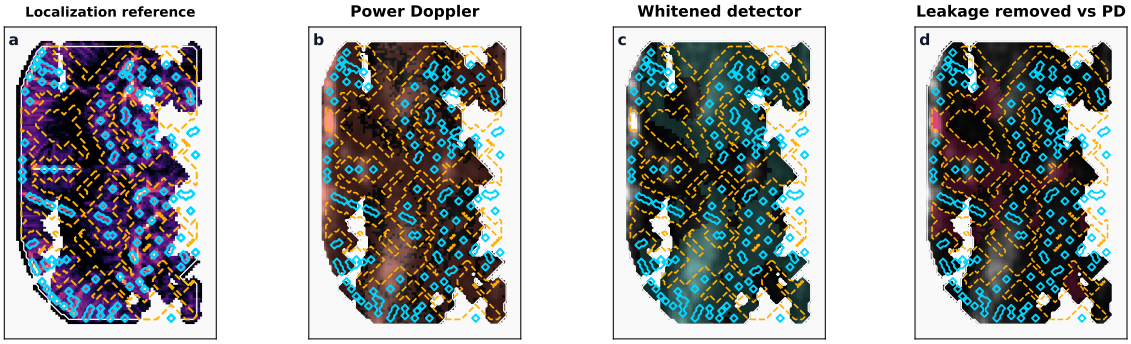

Figure 1: *Illustrative real-IQ structural evaluation preview.* Same audited 64-frame residual window shown with a localization-derived vascular reference, baseline power Doppler, and the fully whitened detector. Panels b–c are thresholded at matched 70% vessel-core recall. Full quantitative same-residual results are reported in Section 5.4; the held-out SIMUS benchmark sets the default detector choice.

We isolate the detector stage after clutter suppression. All detector comparisons use the same clutter-filtered residual so detector effects can be separated from clutter-filter effects.

We evaluate the detector on held-out structural benchmarks, clutter-stress simulations, a structurally labeled phantom, and open real-IQ audits.

What collaboration looks like is straightforward. Labs with raw complex IQ or beamformed slow-time data, acquisition metadata, and task timing can evaluate this detector as a drop-in replacement for the final PD/Kasai-style readout within an existing clutter-filtered pipeline. We run same-residual comparisons against the lab's current readout and return structural or activation analyses in the lab's existing downstream workflow, together with the detector implementation and reusable evaluation harness.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter or motion, and n is residual noise.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. Physically, P_f is the Doppler band where the targeted microvascular flow is expected, P_g is a transition guard band, and P_a captures aliased high-velocity flow or vibration. Band-discretization details and sentinels are documented in Supplementary Material.

2.1 Related context

Spatiotemporal SVD, RPCA, HOSVD, local adaptive clutter filtering, and learned clutter filters are established ways to improve the residual itself in ultrafast Doppler and fUS [11, 12, 31, 32, 45]. The question here is narrower: once a clutter-filtered slow-time cube has been fixed, can the downstream detection statistic reduce artifact leakage more effectively than PD- or Kasai-style readouts on that same residual? That separates this paper from Yu-, Baranger-, and Song-style methods, which adapt the clutter filter itself before a conventional Doppler score is applied [14, 15, 17].

2.2 Baseline scores and detector definitions

Baseline residual and baseline score. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). A baseline clutter-suppression operator (e.g. motion-compensated SVD) produces a residual cube $X_{\text{base}}[t, y, x]$, from which we define the baseline ROC score map

$$S_{\text{base}}(y, x) := \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 \,. \tag{1}$$

This is ordinary power Doppler on X_{base}; short-ensemble sections additionally compare against Kasai lag-1 magnitude and related conventional readouts [16]. S_{base} denotes the baseline ROC score map on X_{base}, and S_{det} the matched-subspace detector score map. Unless otherwise stated, the detector family consists of a fixed statistic, a prespecified adaptive statistic, and a fully whitened covariance-adaptive variant.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final detection statistic rather than replacing upstream clutter suppression. After forming a clutter-filtered slow-time residual, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The detector family has three variants: a fixed statistic, a prespecified adaptive statistic that whitens only clutter-dominant neighborhoods, and a fully whitened variant for settings where stronger local adaptation is beneficial.

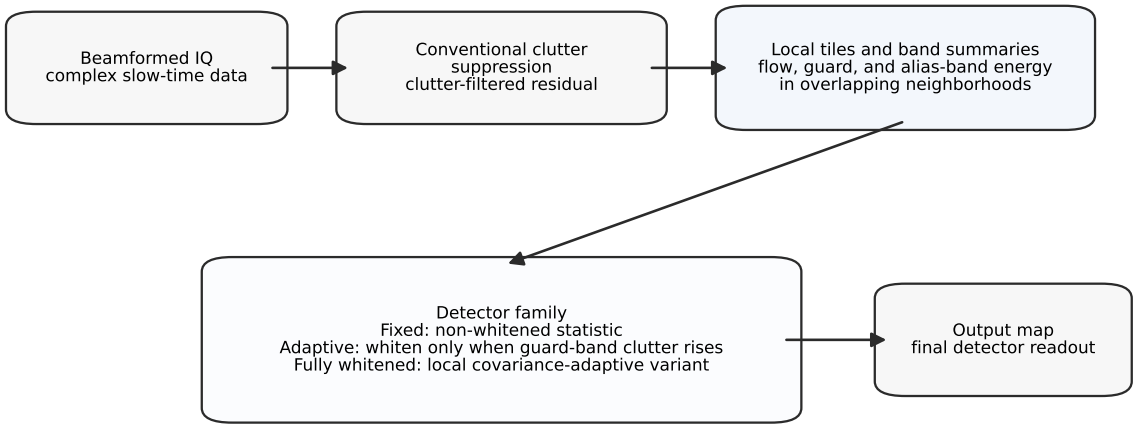

Figure 2: Method overview. Beamformed IQ first undergoes ordinary clutter filtering, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere.

Plain-language interpretation. The fixed statistic is a local band-limited flow detector applied after ordinary clutter filtering. In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow or more like locally structured nuisance? The adaptive branch adds tile-local whitening only where guard-band energy indicates localized clutter. Figure 2 summarizes the pipeline.

Three detector variants. On tile i, let $x_{i,k}$ denote the unwhitened slow-time snapshots and $y_{i,k} = R_i^{-1/2} x_{i,k}$ the whitened snapshots when whitening is active. The fixed detector is

$$S_0(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{x_{i,k}^{\mathrm{H}} P_f \, x_{i,k}}{x_{i,k}^{\mathrm{H}} (I - P_f) \, x_{i,k} + \rho \, x_{i,k}^{\mathrm{H}} x_{i,k} + \epsilon}, \tag{2}$$

which omits local whitening.

$$S_w(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon}. \tag{3}$$

The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\mathrm{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\mathrm{base}}(i) < \tau_g, \end{cases} \tag{4}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once for the setting. The fully whitened detector uses $S_w(i)$ everywhere.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, and adaptive threshold) are fixed for each setting before held-out results are reported. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \text{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\text{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on setting). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. These operating points are stress tests for rare false detections, not proposed clinical thresholds. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 6, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report TPR(α) at the intended operating point; in labeled settings, the negative set is available for *exact* calibration, while in deployment it would be estimated via a negative-only training procedure or from a fixed calibration set. The Results section therefore includes two non-retrospective transfer checks: a fixed-calibration audit on SIMUS-Struct and a complementary block-split fixed-calibration audit on the ULM structural evaluation.

How the prespecified defaults were chosen. Band edges are tied to acquisition physics, while tile geometry and L_t are chosen once per setting to preserve spatial adaptation with stable detector estimates and then held fixed. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to keep the fixed score in cleaner settings and enable whitening only when guard-band clutter evidence is clearly non-negligible.
Additional checks for acquisition metadata and compute gating are deterministic and are applied before reporting.

4.2 Regimes and scope

The evidence is stratified by setting so that quantitative detection results remain clearly separated from robustness or transfer audits. We use the short labels kWave-Stress, Gammex-Struct, SIMUS-Struct-Mobile, SIMUS-Struct-Intraop, ULM-CoreShell, ULM-Motion, and Shin-ProxyMotion throughout; SIMUS-Struct denotes the paired mobile and intra-operative structural benchmark. Downstream detection-statistic design matters after a fixed clutter filter; the fixed band-limited statistic is the default structural choice on the held-out SIMUS benchmark; adaptive

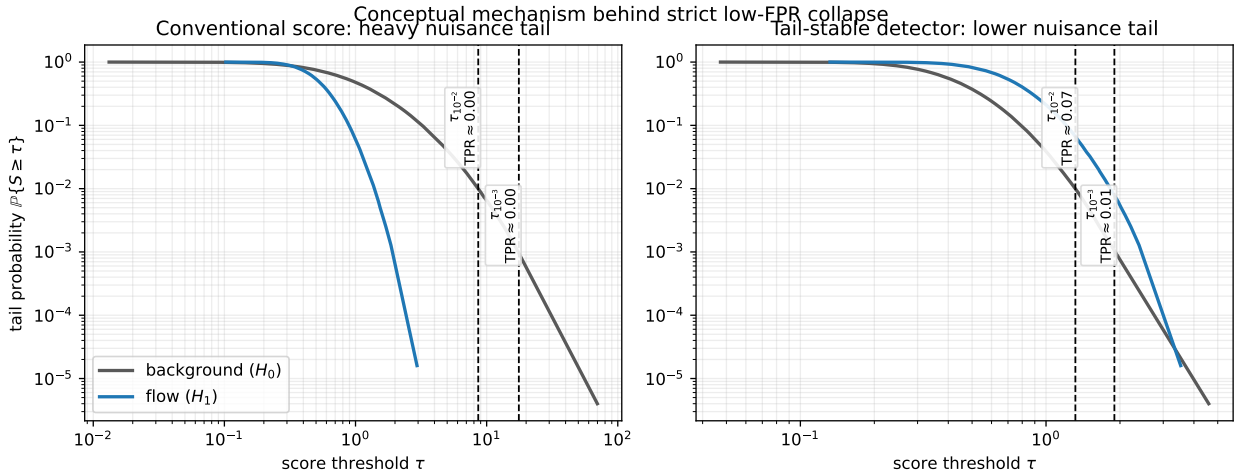

Figure 3: Conceptual strict-tail mechanism. Left: a conventional score can still show moderate-FPR separation while failing at a stricter operating point because the background right tail pushes the background-calibrated threshold $\tau_{10^{-3}}$ above essentially all flow pixels, yielding TPR ≈ 0 by construction. Right: when the nuisance tail is reduced, the same strict background allowance leaves some flow pixels above threshold. The later labeled-brain and Gammex tail figures are dataset-specific examples of this same mechanism.

and fully whitened variants help in narrower clutter-heavy or structure-concentration settings; and fixed-threshold transfer remains limited and setting-dependent.

Each setting uses one frozen operating profile for baseline, detector support, and Doppler-band geometry.

Equation (1) defines the baseline ROC score, Equation (2) the fixed detector, Equation (4) the adaptive detector, and Equation (3) the fully whitened detector. Setting-specific whitening estimators, loading values, and training-support policies are fixed before reporting.

5 Results

5.1 SIMUS-Struct benchmark

SIMUS-Struct uses the open-source SIMUS/PyMUST moving-scatterer forward model and is calibrated to reported fUS operating conditions rather than to a generic parameter sweep [8,9,35,46–48]. We report two prespecified structural settings, SIMUS-Struct-Mobile and SIMUS-Struct-Intraop, with frozen positive, background, and nuisance masks and held-out evaluation on seeds 127–128 after development on seeds 125–126. The positive class is unaliased parenchymal Pf-band flow; the negatives are ordinary background and explicit nuisance structures such as superficial high-velocity or specular components. Throughout this section, $AUC_{\text{main/bg}}$ measures separation from background, $AUC_{\text{main/nuis}}$ measures separation from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ measures nuisance leakage when half of the target positives are retained.

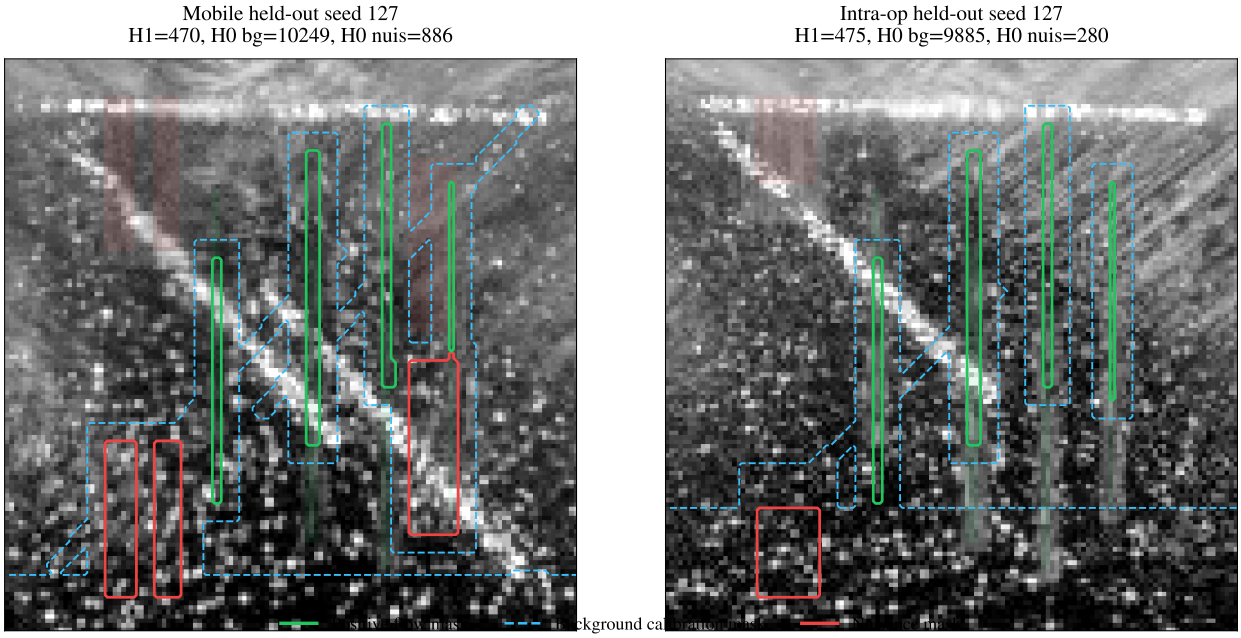

Figure 4: Representative SIMUS-Struct geometry for the SIMUS-Struct-Mobile and SIMUS-Struct-Intraop reference settings. The grayscale underlay is log beamformed power from the simulated IQ cube and is shown only to make the mask geometry visible. Green contours mark the parenchymal-flow positive set H_1, blue dashed contours mark the broad background pool H_0^{bg} used to set the score threshold, and red contours mark the nuisance-negative set H_0^{nuis} used to quantify structured false detections at that same operating point. Table 2 reports the held-out seed results on these frozen mask types across seeds 127–128.

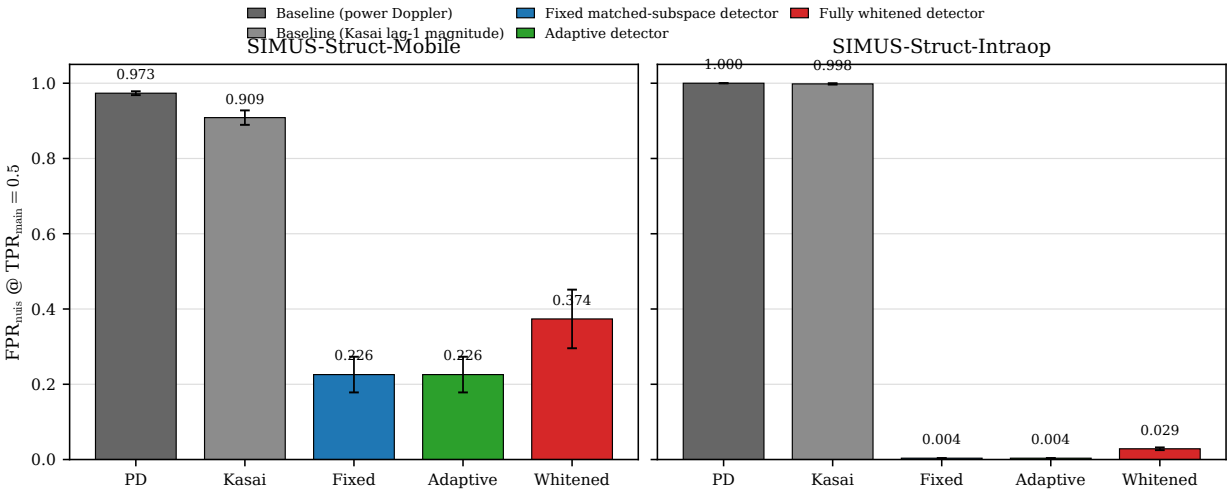

Figure 5: Headline same-residual scoring-rule comparison on the held-out SIMUS reference benchmark using the common MC–SVD residual from Table 3. Bars show nuisance false-positive rate at matched main-response recall 0.5, the clearest operating-point summary of the benchmark. Lower bars are better. On the intra-operative setting, the fixed matched-subspace detector reduces nuisance FPR from 0.998 for Kasai to 0.004 on the same residual stream; under this frozen guard rule, the adaptive variant matches that result. $\mathrm{AUC}_{\mathrm{main/nuis}}$ values are reported in Table 3.

Same-residual scoring-rule ablation. We first fix one held-out MC–SVD residual stream and swap only the downstream detection statistic to isolate the detector effect directly. This removes clutter-filter choice from the comparison. On that fixed residual, PD and Kasai still rank nuisance highly, the fixed matched-subspace statistic is the strongest same-residual detector, and the fully whitened statistic is worse on both held-out settings. The adaptive statistic is numerically identical to the fixed statistic here because its frozen guard-promotion rule does not activate on these cleaner MC–SVD residuals. The adaptive extension therefore preserves the fixed default when the clutter evidence is weak instead of forcing whitening where it does not help. The full same-residual comparison is reported in Table 3 below.
Table 2 is the best-chain benchmark: each row reports the strongest held-out end-to-end pipeline for that setting, so both clutter filter and final detector may differ across rows.

Best-chain benchmark. We then report the strongest end-to-end chain in each held-out setting. On the mobile setting, HOSVD → Kasai is the strongest conventional baseline. Adaptive Global SVD → fixed matched-subspace improves $\mathrm{AUC}_{\mathrm{main/nuis}}$ from 0.672 [0.624, 0.720] to 0.953 [0.923, 0.984] and yields zero observed nuisance detections on both held-out seeds. On the intra-operative parenchymal setting, HOSVD → Kasai is again the strongest conventional baseline, while RPCA → fixed matched-subspace improves $\mathrm{AUC}_{\mathrm{main/nuis}}$ from 0.451 [0.431, 0.471] to 0.99996 [0.99994, 0.99998] and again yields zero observed nuisance detections on both held-out seeds. Figure 4 shows the held-out mask geometry used for these comparisons, and the 0.000 nuisance rows in Table 2 are literal zero observed nuisance detections rather than rounded near-zero rates. The detector-only same-residual comparison is reported separately in Table 3 and Figure 5. This benchmark sets the default detector configuration.

Mobile

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: Adaptive Global SVD → fixed matched-subspace.

Summary	Best conventional baseline			Fixed detector chain		
	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
Mean [min,max] over seeds 127–128	0.669 [0.647,0.692]	0.672 [0.624,0.720]	0.286 [0.210,0.361]	0.851 [0.798,0.905]	0.953 [0.923,0.984]	$0.000^{\dagger}$ [0.000,0.000]

Intra-operative parenchymal

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: RPCA → fixed matched-subspace.

Summary	Best conventional baseline			Fixed detector chain		
	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
Mean [min,max] over seeds 127–128	0.837 [0.834,0.839]	0.451 [0.431,0.471]	0.621 [0.600,0.643]	0.994 [0.993,0.995]	0.99996 [0.99994,0.99998]	$0.000^{\dagger}$ [0.000,0.000]

†Exactly zero observed nuisance detections; supported floor 1/886 (mobile) and 1/280 (intra-operative parenchymal).

Table 2: Held-out SIMUS-Struct reference settings. For each evaluated structural setting, we report the best conventional baseline from the development-split baseline sweep and the strongest fixed detector chain chosen on development seeds 125–126, then evaluated unchanged on held-out seeds 127–128. Values are mean [min,max] over the two held-out seeds. $\mathrm{AUC}_{\mathrm{main/bg}}$ measures separation of target flow from ordinary background, $\mathrm{AUC}_{\mathrm{main/nuis}}$ measures separation of target flow from explicit nuisance structures, and $\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives.

Method	Mobile			Intra-operative parenchymal		
	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ @ $\mathrm{TPR}_{\mathrm{main}} = 0.5$
Baseline (power Doppler)	0.568 [0.562,0.575]	0.131 [0.127,0.134]	0.973 [0.968,0.979]	0.912 [0.909,0.915]	0.022 [0.016,0.028]	1.000 [1.000,1.000]
Baseline (Kasai lag-1 magnitude)	0.500 [0.489,0.512]	0.171 [0.154,0.188]	0.909 [0.889,0.928]	0.792 [0.788,0.795]	0.019 [0.017,0.022]	0.998 [0.996,1.000]
Fixed matched-subspace detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Adaptive detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Fully whitened detector	0.569 [0.553,0.585]	0.600 [0.529,0.672]	0.374 [0.296,0.451]	0.819 [0.819,0.820]	0.932 [0.927,0.936]	0.029 [0.025,0.032]

Table 3: Same-residual detector-family ablation on SIMUS-Struct using a common MC–SVD residual. Rows differ only in the downstream scoring rule: power Doppler, Kasai lag-1 magnitude, the fixed flow-band matched-subspace detector without whitening ($R = I$), the adaptive detector that promotes clutter-heavy tiles onto a whitened branch, and the fully whitened detector. Values are means with [min,max] over held-out evaluation seeds 127 and 128 for each setting. Lower nuisance FPR at matched TPR is better.

FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ directly measures artifact leakage at a clinically relevant sensitivity operating point.

A fixed-threshold transfer audit also survives on the same benchmark family: on SIMUS-Struct-Mobile, the fixed statistic preserves nonzero held-out main recall while reducing realized nuisance FPR relative to PD, and on SIMUS-Struct-Intraop it is the only statistic that suppresses nuisance leakage to zero under the transferred threshold.

Clutter-heavy stress setting. Whitening starts to matter once the same benchmark is pushed into clutter-heavier settings. We test two held-out stresses: stronger cardiac-like low-frequency pulsation and shorter slow-time support (64, 48, and 32 frames). The pulsation rows use a fixed 1.30 Hz nuisance term with increasing amplitude, and the short-ensemble rows mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow

burden [1, 11, 12, 23, 35]. The held-out evaluation is then rerun at full reporting tier for the clutter-filter families that survived the reduced-grid search.

Cardiac-like pulsation

Metric	Baseline (RPCA→PD)	Fixed statistic	Adaptive statistic	Whitened variant
Ref. 1.0x cardiac amplitude at 1.30 Hz				
$AUC_{main/nuis}$	0.381 [0.345, 0.416]	0.879 [0.860, 0.898]	0.879 [0.860, 0.898]	0.911 [0.894, 0.925]
FPR_{nuis}@0.5	0.743 [0.652, 0.842]	0.034 [0.023, 0.047]	0.034 [0.023, 0.046]	0.017 [0.008, 0.027]
Mod. 2.0x cardiac amplitude at 1.30 Hz				
$AUC_{main/nuis}$	0.388 [0.354, 0.423]	0.877 [0.857, 0.895]	0.877 [0.857, 0.895]	0.903 [0.886, 0.920]
FPR_{nuis}@0.5	0.668 [0.593, 0.795]	0.049 [0.030, 0.064]	0.049 [0.030, 0.064]	0.030 [0.019, 0.043]
Hard 3.0x cardiac amplitude at 1.30 Hz				
$AUC_{main/nuis}$	0.399 [0.365, 0.433]	0.875 [0.854, 0.893]	0.875 [0.854, 0.894]	0.905 [0.890, 0.921]
FPR_{nuis}@0.5	0.609 [0.537, 0.770]	0.044 [0.029, 0.058]	0.044 [0.029, 0.059]	0.027 [0.017, 0.041]

Short ensemble

Metric	Baseline (RPCA→PD)	Fixed statistic	Adaptive statistic	Whitened variant
Ref. 64 frames (42.7 ms)				
$AUC_{main/nuis}$	0.381 [0.345, 0.415]	0.879 [0.861, 0.897]	0.879 [0.859, 0.898]	0.911 [0.895, 0.926]
FPR_{nuis}@0.5	0.743 [0.649, 0.840]	0.034 [0.023, 0.046]	0.034 [0.023, 0.047]	0.017 [0.008, 0.027]
Mod. 48 frames (32.0 ms)				
$AUC_{main/nuis}$	0.390 [0.356, 0.423]	0.784 [0.758, 0.807]	0.784 [0.759, 0.808]	0.846 [0.824, 0.868]
FPR_{nuis}@0.5	0.754 [0.708, 0.814]	0.137 [0.102, 0.169]	0.137 [0.103, 0.167]	0.058 [0.034, 0.080]
Hard 32 frames (21.3 ms)				
$AUC_{main/nuis}$	0.426 [0.392, 0.462]	0.772 [0.746, 0.796]	0.772 [0.746, 0.797]	0.716 [0.686, 0.745]
FPR_{nuis}@0.5	0.617 [0.536, 0.727]	0.166 [0.139, 0.194]	0.166 [0.139, 0.193]	0.160 [0.130, 0.207]

Table 4: Held-out SIMUS mobile stress setting with RPCA fixed as the clutter filter on every row. Entries report $AUC_{main/nuis}$ and FPR_{nuis}@0.5, with point estimates shown as x [lo, hi] using 95% nonparametric bootstrap intervals over the held-out masked score samples. RPCA is used because RPCA → PD is the strongest conventional baseline among evaluated methods on every reported stress row. This table separates the fixed, adaptive, and fully whitened detector statistics directly instead of reporting only the best family member on each row.

Table 4 shows the punchline directly: the fully whitened statistic is strongest on the cardiac-pulsation rows and on the 64-frame and 48-frame short-ensemble rows, while the fixed statistic

remains strongest only on the hardest 32-frame row. The adaptive statistic is numerically identical to the fixed statistic on these rows, so the gain is not coming from freely switching among all three modes. Because this stress setting uses one held-out evaluation seed per axis, the uncertainty notation here is a within-seed bootstrap interval over the held-out masked scores rather than a between-seed interval.

Functional same-cube detection-statistic comparison. A held-out functional audit on the same residual stream is directionally positive for the fully whitened variant across most residual families, but the effect is smaller than in the structural benchmark and is treated as secondary evidence.

5.2 Brain simulation stress tests

The held-out SIMUS-Struct benchmark already determines the default downstream statistic, so the labeled brain simulations are kept brief here. These controlled kWave-Stress audits use labeled brain simulations generated with the k-Wave wave-propagation toolbox [13] and ask a narrower mechanism question on the same MC–SVD residual cube: does the gain come from band selectivity, from whitening alone, or from their combination? The answer is that whitening alone is not sufficient in these stress tests. On fixed residuals, whitened total power collapses at the reported strict-tail operating points, whereas the fixed band-limited matched-subspace score remains strong; detector-swap audits also show that gains do not automatically persist on more aggressive clutter filters and that the fully whitened variant remains stable under modest covariance-training trim changes. Taken together, these labeled stress tests explain why the held-out SIMUS benchmark favors the fixed statistic and why whitening becomes useful only in narrower clutter-heavy settings.

5.3 Gammex phantom: structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [43], which provides complex beamformer output from phantom acquisitions on a clinical scanner. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Sequence-level PRF values and decoder/QC details are fixed once per family and documented in Supplementary Material; the main text uses only the reported short-ensemble profile needed for the structural comparison below.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler or matched-subspace outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots), we use one frozen short-ensemble profile: baseline SVD band-pass keep range $[2, 17]$, detector aperture $L_t = 16$, tiles 8×8, Tyler-type covariance estimation, diagonal loading $\lambda = 0.07$, and fixed band definitions $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. The along- and across-view stride choices are fixed from geometry alone and documented in Supplementary Material. For baseline context on the same residual stream, we also report power Doppler, log power Doppler, and Kasai lag-1 power.

Structural-label operating points (Gammex). The same strict-tail geometry of Figure 3 appears in the short-ensemble Gammex phantom, where residual twinkling structure dominates the upper background tail. We therefore pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 5 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three detector-family score variants evaluated on that identical residual cube (whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. On the short-ensemble Gammex phantom, changing only the downstream score recovers lumen-consistent structure from the same residual. On the representative Figure 6 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the matched-subspace score recovers 586 lumen pixels while holding the same background allowance. Local whitening is the dominant ingredient in this setting, and whitened power outperforms the matched-subspace ratio on both views, especially across-view. Gammex therefore supports localized covariance-aware downstream scoring, but it does not make the matched-subspace ratio the preferred phantom score under short-ensemble conditions. The across view provides the clearest structural-label evidence, whereas the along view remains a harsher artifact-tail stress test under this profile; a small untuned sanity audit in Supplementary Material gives similar values and leaves the qualitative ordering unchanged.
Additional Gammex sanity checks are provided in the Supplementary Material.

Within-ensemble motion ladders. A labeled within-ensemble motion ladder on the along-view Gammex cine is directionally consistent with the main phantom result and is useful as a robustness check, but the central phantom claim in the main text is the same-residual structural ROC comparison above.

5.4 ULM real-IQ audits

We next test the detector on open real-IQ rat-brain data using conservative structural endpoints.

Along view (linear17 @ PRF 2500)				
Method / score	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Power Doppler	0.00000	0.00000	0.00000	0.00000
Kasai	0.00000	0.00000	0.00000	0.00000
Whitened power	**0.4116**	**0.0012**	**0.0119**	**0.0532**
Unwhitened ratio	0.00031	0.00000	0.00000	0.00001
Fully whitened detector	0.0942	0.00000	0.00003	0.00031
Across view (linear17 @ PRF 2500)				
Method / score	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Power Doppler	0.00000	0.00000	0.00000	0.00000
Kasai	0.00000	0.00000	0.00000	0.00000
Whitened power	**0.7234**	**0.4820**	**0.5155**	**0.5813**
Unwhitened ratio	0.1095	0.0574	0.0602	0.0653
Fully whitened detector	0.6891	0.0491	0.0998	0.2237

Table 5: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Every row is evaluated on the identical baseline SVD band-pass residual cube; only the final scoring rule changes. Rows compare power Doppler, Kasai, whitened power, an unwhitened matched-subspace ratio, and the fully whitened matched-subspace detector on that same residual. The table includes $\alpha = 10^{-2}$ as a practitioner-facing relaxed operating point together with the stricter $\alpha \leq 10^{-3}$ tail regime used for stress testing. Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Point estimates are shown here for readability; full 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) are reported in the Supplementary Material. Pooled background counts are $n_{\text{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\text{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Best TPR in each column is bolded. Reproducibility details are provided in Supplementary Material.

Dataset and loader. We use the public kHz IQ ULM benchmark released for localization microscopy benchmarking [38, 39]. For the structural evaluation reported here, a single MC–SVD baseline profile is frozen in advance and then held fixed; loader details and the one-time label-free baseline sweep are provided in Supplementary Material.

5.4.1 Localization-derived structural evaluation

ULM-CoreShell. Using a published PALA localization rendering as an external vascular reference, we ask whether detector scores concentrate on vessel cores rather than the adjacent parenchyma. The reference comes from the published PALA rat-brain localization rendering, registered once back to the IQ grid and converted into a conservative vessel-core versus perivascular-shell endpoint [38,39]. That registration is part of label construction only; the detector itself is evaluated on frozen 64-frame windows without frame-to-frame registration. Additional registration, overlap, and falsification checks are reported in Supplementary Material.

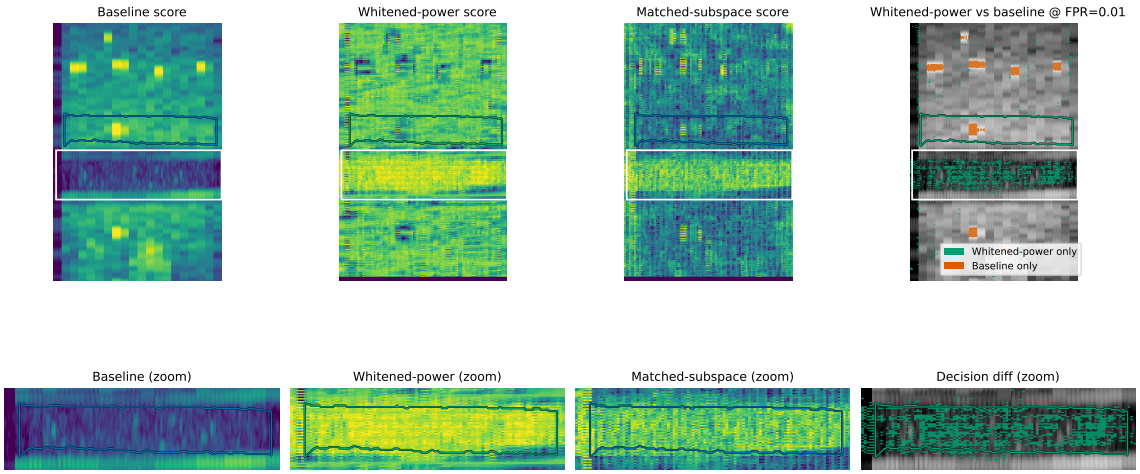

Figure 6: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline power Doppler, whitened power, matched-subspace, and a matched-FPR decision-difference panel at background false-positive rate 10^{-2}, showing pixels detected only by whitened power versus only by baseline. Bottom row: zoomed region of interest. The tube contour is overlaid for reference. The qualitative ordering matches Table 5: whitened power is the strongest phantom score in this short-ensemble setting, while the matched-subspace panel remains included as the detector-core reference.

Score	AUC (H1 vs H0)	TPR@10^{-2}	TPR@10^{-3}	shell FPR @70% core recall
Baseline (power Doppler)	0.504 [0.498,0.510]	0.010 [0.009,0.011]	7.9e-5 [0,2.4e-4]	0.698 [0.689,0.706]
Baseline (Kasai lag-1 magnitude)	0.503 [0.496,0.509]	0.009 [0.008,0.010]	5.9e-5 [0,1.8e-4]	0.701 [0.693,0.709]
Fixed matched-subspace statistic	0.514 [0.508,0.521]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.703 [0.694,0.711]
Adaptive statistic	0.514 [0.507,0.520]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.702 [0.693,0.711]
Fully whitened variant	**0.530 [0.524,0.537]**	**0.024 [0.021,0.027]**	**0.004 [0.003,0.006]**	**0.676 [0.665,0.687]**

Table 6: Same-residual ULM structural audit on real in vivo rat-brain IQ. All rows use the same frozen 64-frame MC–SVD residual cube and the same localization-derived vessel-core versus shell masks; only the downstream scoring rule changes. Entries report window-level means with 95% bootstrap intervals over the 70 audited windows. Best values in each metric column are bolded. The fully whitened variant exceeds baseline power Doppler on all 10 audited blocks (exact paired sign-flip test on block-mean AUC, p = 0.002).

On this endpoint, Table 6 reports PD, Kasai, the fixed statistic, the adaptive statistic, and the fully whitened variant on the same residual windows and the same localization-derived vessel-core versus shell masks. The fixed and adaptive statistics yield only modest pooled-ROC gains over PD and Kasai, and adaptive is nearly indistinguishable from fixed under this profile. The fully whitened variant is the only statistic that improves pooled AUC, strict-tail sensitivity, and shell FPR at matched 70% vessel-core recall: AUC 0.530 [0.524, 0.537], TPR@10^{-3} 0.004 [0.003, 0.006], and shell FPR 0.676 [0.665, 0.687], compared with AUC 0.514/0.514, TPR@10^{-3} 2.6e-4/2.6e-4, and shell FPR 0.703/0.702 for the fixed/adaptive statistics and AUC 0.504/0.503, TPR@10^{-3} 7.9e-5/5.9e-5, and shell FPR 0.698/0.701 for PD/Kasai.

The pooled advantage is not driven by one favorable block. Relative to the fixed statistic, the fully whitened variant exceeds AUC on 47/70 audited windows and all 10/10 blocks, lowers

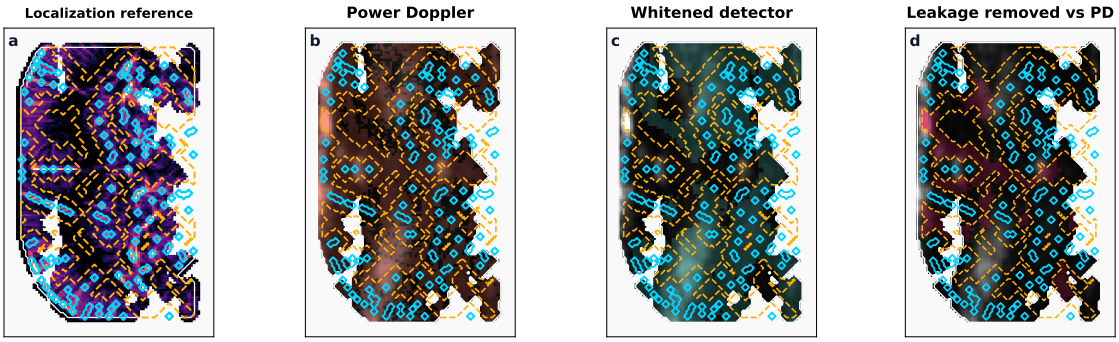

Figure 7: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace variant. Table 6 reports the formal same clutter-filtered-residual comparison of detection statistics.

shell FPR at matched 70% core recall on 47/70 windows and 9/10 blocks, and yields nonzero TPR@10^{-3} in 45/70 windows versus 10/70 for the fixed and adaptive statistics. Against baseline power Doppler, the fully whitened variant also wins AUC on all 10 audited blocks on this same-acquisition ULM structural evaluation, and an exact paired sign-flip test on the block-mean AUC difference gives $p = 0.00195$. Figure 7 gives the visual counterpart to this result. On an RTX 4080 SUPER, the same 64-frame no-registration configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget under the reported replay conditions.

Supplementary overlap and falsification checks point in the same direction: a stricter odd-block-reference / even-block-evaluation split preserves the family ordering with attenuated margins, while support-matched randomization and large cyclic translations collapse the structural endpoint toward chance rather than reproducing the fully whitened ordering.

A fixed-threshold audit on the same structural endpoint shows that PD and Kasai collapse to zero held-out vessel-core recall, whereas the fully whitened variant retains a small but nonzero held-out core TPR under a transferred shell-calibrated threshold. That result remains modest, but it shows the structural claim does not depend entirely on per-window threshold readjustment.

5.4.2 Motion robustness audit

Motion audit. A motion-injection audit on ULM-CoreShell and Shin-ProxyMotion provides secondary robustness evidence under controlled residual-motion stress. In those audits, matched-subspace score maps are more stable than the baseline maps, but these results remain corroboration rather than the structural efficacy claim that anchors the main real-IQ section.

Shin-ProxyMotion. To avoid making the motion-robustness story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the

supplement [36,37]. Under a prespecified low-frequency operating setting and a controlled residual-motion stress test, the matched-subspace map is more stable than the MC–SVD baseline map across all three audited windows, while the separate fixed-threshold proxy-transfer check remains negative.

Replay feasibility. Under the reported steady-state replay conditions, the final detector configurations remain below the acquisition-time budget on the labeled-brain, Shin, and Gammex evaluations.

6 Closing perspective

Post-beamformed fUS benefits from a better downstream detection statistic, not only from a better clutter filter. On the held-out same-residual SIMUS benchmark, the fixed statistic lowers intra-operative nuisance FPR from 0.998 for Kasai to 0.004 at matched 50% target-flow recall, while the separate best-chain benchmark yields zero observed nuisance detections on all four held-out seed-setting pairs. The mechanism is setting-dependent: the fixed band-limited statistic is the structural default, while stronger whitening helps in narrower clutter-dominant settings.

The open real-IQ sections provide conservative structural corroboration on real rat-brain IQ and favor the fully whitened variant, but they are not yet a task-based functional validation. In practice, the fixed statistic is the default; the fully whitened variant is reserved for persistently clutter-dominant windows when the latency budget allows it.

Collaboration opportunities

The clearest next test is prospective validation on real task-driven fUS data. A collaborator would provide raw complex IQ or beamformed slow-time data, PRF and acquisition metadata, stimulus or task timing, and, when available, an independent structural anchor such as a localization-derived vascular map or structural co-registration. We would run the detector family as a drop-in replacement for the final PD/Kasai-style readout on the collaborator's existing clutter-filtered residual pipeline, produce same-residual comparisons against PD/Kasai-style scores, and evaluate task-activation or structural readouts with the collaborator's current downstream analysis workflow rather than replacing it. The intended output is a joint validation study, positive or negative, together with a reusable evaluation harness the collaborator can continue using on future datasets. Collaborators would also receive the full detector implementation configured for their acquisition parameters, together with the evaluation harness, for use on future datasets. The contribution from this side is the tested detector implementation, the same-residual evaluation protocol, and the computational work needed to run the comparisons and package the results. The author welcomes collaboration inquiries from labs with raw fUS IQ data and task paradigms; correspondence to arthur@skymesasystems.com.

Data and code availability

All datasets used in this work are publicly available or derived deterministically from those sources. The reference implementation, reproduction scripts, and canonical commit information are available at `https://github.com/ArthurShune/fus-detectors`; the repository is released under the

MIT License and is the reference implementation used for the reported analyses (Supplementary Material).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Treeby, B. E., and Cox, B. T. k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields. *Journal of Biomedical Optics* 15(2), 021314 (2010). DOI: `https://doi.org/10.1117/1.3360308`.

[14] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: https://doi.org/10.1109/TUFFC.2010.1521.

[15] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: https://doi.org/10.1109/TMI.2016.2605819.

[16] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: https://doi.org/10.1109/T-SU.1985.31615.

[17] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: https://doi.org/10.1109/TUFFC.2016.2606598.

[18] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: https://doi.org/10.1016/j.ultras.2020.106163.

[19] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[20] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[21] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: https://doi.org/10.1126/scitranslmed.aah6756.

[22] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: https://doi.org/10.1038/s41467-021-21387-x.

[23] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: https://doi.org/10.1126/sciadv.adu9133.

[24] Lambert, T., Brunner, C., Urban, A., et al. Functional ultrasound imaging and neuronal activity: How accurate is the spatiotemporal match? *Imaging Neuroscience* 3, IMAG.a.34 (2025). DOI: https://doi.org/10.1162/IMAG.a.34.

[25] Edelman, B. J., et al. The COMBO window: A chronic cranial implant for multiscale circuit interrogation in mice. *PLOS Biology* 22(6), e3002664 (2024). DOI: https://doi.org/10.1371/journal.pbio.3002664.

[26] Chen, Z., et al. Functional ultrasound imaging of auditory responses in comatose patients. *Research* 8, 0709 (2025). DOI: https://doi.org/10.34133/research.0709.

[27] Nunez-Elizalde, A. O., et al. Neural correlates of blood flow measured by ultrasound. *Neuron* 110(10), 1631–1640.e4 (2022). DOI: https://doi.org/10.1016/j.neuron.2022.02.012.

[28] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[29] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[30] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[31] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[32] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: `https://doi.org/10.1109/TMI.2019.2941271`.

[33] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[34] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[35] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: `https://doi.org/10.3389/fnins.2019.01384`.

[36] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[37] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: `https://doi.org/10.1038/s41467-024-47154-2`.

[38] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[39] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: `https://doi.org/10.1038/s41551-021-00824-8`.

[40] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: https://doi.org/10.1038/nature16066.

[41] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: https://doi.org/10.1016/j.ultras.2017.01.008.

[42] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: https://doi.org/10.1109/TUFFC.2022.3211469.

[43] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[44] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: https://doi.org/10.1109/TUFFC.2021.3073292.

[45] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[46] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106726.

[47] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106774.

[48] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[49] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[50] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: https://doi.org/10.1002/j.1538-7305.1978.tb02104.x.

[51] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: https://doi.org/10.1109/PROC.1982.12433.

[52] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques.* Cambridge University Press (1993).

[53] Guerci, J. R. Artech House monograph (2003).

[54] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: `https://doi.org/10.1109/MSP.2006.1593337`.

[55] Melvin, W. L., and Guerci, J. R. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248215`.

[56] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: `https://doi.org/10.1109/TAES.2006.248209`.

[57] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: `https://doi.org/10.1109/53.665`.

[58] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[59] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: `https://doi.org/10.1109/36.957286`.

[60] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[61] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[62] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[63] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[64] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[65] Ward, J. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).